

ADVANCED QUANTUM MECHANICS

Quantum many body physics



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LECTURE 14: The Helium atom

We now turn to a genuinely many-body Coulomb problem with two indistinguishable electrons (including spin). This is the [helium atom](#). This element has many [applications](#).

We proceed as follows:

- write the full three-body Hamiltonian (nucleus + two electrons),
- place the center of mass at the origin (the nucleus is essentially fixed),
- obtain the electronic Hamiltonian in the coordinates $\mathbf{r}_1, \mathbf{r}_2$,
- identify the central difficulty: the electron–electron repulsion couples the motions and prevents exact separation,
- construct a variational ansatz using hydrogenic $1s$ orbitals with an effective nuclear charge Z_{eff} ,
- compute the expectation value of the full Hamiltonian in this trial state,
- minimize with respect to Z_{eff} to obtain an approximate ground-state energy close to the exact value.

Two-electron Coulomb Hamiltonian

We consider the helium atom: a nucleus of charge $+2e$ and mass M_{He} (approximately the mass of the He^{2+} ion) and two electrons of mass m and charge $-e$. We will simplify our lives from the start and will assume the nuclei very heavy so we neglect the nuclear problem fully being left with only the two electron problem.

Let $\mathbf{r}_1$ and $\mathbf{r}_2$ be the positions of the electrons, and define the electron–electron separation

$$r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|.$$

The exact electronic Hamiltonian is $\hat{H} = \hat{H}_{\text{el}} \otimes \mathbb{1}_{\text{spin}}$ where the electronic part is

$$\hat{H}_{\text{el}} = \hat{h} \otimes \mathbb{1} + \mathbb{1} \otimes \hat{h} + \hat{V}_{12}.$$

with the single particle Hamiltonian is

$$\hat{h} = -\frac{\hbar^2}{2m_e} \nabla_{\mathbf{r}}^2 - \frac{1}{4\pi\epsilon_0 r}$$

and the electron–electron Coulomb interaction term is

$$\hat{V}_{12} = \frac{e^2}{4\pi\epsilon_0 r_{12}}$$

couples the motion of the two electrons and prevents exact separation of variables. The stationary Schrödinger equation for the electronic wavefunction $\Psi(\mathbf{r}_1, \mathbf{r}_2)$ reads

$$\hat{H}_{\text{el}}\Psi(\mathbf{r}_1, \mathbf{r}_2) = E \Psi(\mathbf{r}_1, \mathbf{r}_2).$$

Taking care of the fermionic nature of electrons -

The usual spatial $1s$ orbital for $Z = 2$ is

$$\phi_{1s}^{(Z=2)}(r) = \frac{(2)^{3/2}}{\sqrt{\pi a_0^3}} e^{-2r/a_0}.$$

To include spin and form a spinor-valued orbital we attach the spin basis vector.

Spin-up electron in $1s$:

$$\Phi_{1s,\uparrow}(\mathbf{r}) = \phi_{1s}(\mathbf{r}) \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Spin-down electron in $1s$:

$$\Phi_{1s,\downarrow}(\mathbf{r}) = \phi_{1s}(\mathbf{r}) \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Each orbital is thus a two-component spinor given by the product of the spatial wavefunction and a two-dimensional spin basis vector. Let

$$\Phi_{\uparrow}(\mathbf{r}) = \phi_{1s}(\mathbf{r}) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \phi_{1s}(\mathbf{r}_1) |\uparrow\rangle, \quad \Phi_{\downarrow}(\mathbf{r}) = \phi_{1s}(\mathbf{r}) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \phi_{1s}(\mathbf{r}_1) |\downarrow\rangle.$$

Then the only antisymmetric two-electron state built from the same spatial orbital is

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}} [\Phi_{\uparrow}(\mathbf{r}_1) \otimes \Phi_{\downarrow}(\mathbf{r}_2) - \Phi_{\downarrow}(\mathbf{r}_1) \otimes \Phi_{\uparrow}(\mathbf{r}_2)].$$

Here the spatial part is identical for both electrons, so the antisymmetry of the total state is entirely carried by the spin part:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \phi_{1s}(\mathbf{r}_1) \phi_{1s}(\mathbf{r}_2) \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle).$$

Comment: para- vs. ortho-helium

The historical terminology “para-helium” and “ortho-helium” distinguishes singlet and triplet spin configurations of the two-electron system. In the ground-state configuration $1s\ 1s$ the spatial wavefunction is symmetric, so the Pauli principle enforces an antisymmetric spin part, leading uniquely to the spin-singlet 1S_0 ground state (para-helium). Triplet (ortho) configurations require two different spatial orbitals in order to form an antisymmetric spatial state; these appear only in excited configurations such as $1s\ 2s$ or $1s\ 2p$. Many of these triplet levels are metastable, since radiative decay to the singlet ground state is spin-forbidden.

The electron–electron term

$$\hat{V}_{12} = \frac{e^2}{4\pi\epsilon_0 r_{12}}$$

couples the motion of the two electrons and prevents exact separation of variables. There is no simple closed-form analytical solution for the ground state. Instead, we construct *approximate* solutions using, e.g., variational or Hartree–Fock methods.

0.1 Hartree–Fock procedure for helium

For the helium ground state we use that both electrons occupy the same spatial orbital and form a spin singlet (para-helium). The Hartree–Fock construction specializes to the following steps:

(i) **Single-orbital ansatz.**

In Hartree–Fock we now keep exactly the same structure, but we do not fix the spatial orbital to be the hydrogenic $1s$. Instead we introduce a general, normalized spatial orbital $\phi(\mathbf{r})$ and attach the same spin basis vectors:

$$\Phi_{\uparrow}(\mathbf{r}) = \phi(\mathbf{r}) \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \Phi_{\downarrow}(\mathbf{r}) = \phi(\mathbf{r}) \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

So the only change compared to the discussion above is the replacement $\phi_{1s}(\mathbf{r}) \rightarrow \phi(\mathbf{r})$, and the goal of Hartree–Fock is precisely to determine this orbital $\phi(\mathbf{r})$ by minimizing the total electronic energy of helium. The antisymmetrized two-electron wavefunction constructed from these two one-electron spinors automatically yields the same singlet structure we already derived such that

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \phi(\mathbf{r}_1) \phi(\mathbf{r}_2) \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle).$$

but now with an unknown spatial orbital that we will determine self-consistently. We can simplify the notation in terms of kets

$$|S\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle), \quad |\phi\rangle = \int d^3r \phi(\mathbf{r}) |\mathbf{r}\rangle.$$

such that

$$|\phi\phi\rangle = |\phi\rangle_1 \otimes |\phi\rangle_2, \quad |\Psi\rangle = |\phi\phi\rangle \otimes |S\rangle.$$

(ii) **Energy functional.**

Given the single-orbital ansatz above, both electrons occupy the same spatial orbital $\phi(\mathbf{r})$ with opposite spins. The total electronic energy in Hartree–Fock is obtained by taking the expectation value of the full two-electron Hamiltonian with this antisymmetrized state. The Hartree–Fock is a variational method: we treat $\phi(\mathbf{r})$ as an unknown function and determine it by minimizing the electronic energy. In other words, we convert the many-electron Schrödinger equation into an equivalent *functional minimization problem* for a single orbital subject to normalization. This allows us to approximate the full interacting problem by a self-consistent one-electron equation. The energy functional is defined as

$$E[\phi] = \langle \Psi | \hat{H}_{\text{el}} | \Psi \rangle = \langle \phi\phi | \hat{H}_{\text{el}} | \phi\phi \rangle \langle S | \hat{\mathbf{1}}_{\text{spin}} | S \rangle = \langle \phi\phi | \hat{H}_{\text{el}} | \phi\phi \rangle,$$

because $\langle S | S \rangle = 1$. The one-particle part is easily derived

$$E_h[\phi] = \langle \phi\phi | (\hat{h} \otimes \mathbb{1} + \mathbb{1} \otimes \hat{h}) | \phi\phi \rangle = 2 \langle \phi | \hat{h} | \phi \rangle.$$

The interaction part is

$$E_{12}[\phi] = \langle \phi\phi | \hat{V}_{12} | \phi\phi \rangle = \iint d^3r_1 d^3r_2 \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|} |\phi(\mathbf{r}_1)|^2 |\phi(\mathbf{r}_2)|^2.$$

This gives the full Hartree–Fock energy functional

$$E[\phi] = 2 \int d^3r \phi^*(\mathbf{r}) \left(-\frac{\hbar^2}{2m} \nabla^2 - \frac{2e^2}{4\pi\epsilon_0 r} \right) \phi(\mathbf{r}) + \iint d^3r_1 d^3r_2 \frac{|\phi(\mathbf{r}_1)|^2 |\phi(\mathbf{r}_2)|^2}{4\pi\epsilon_0 r_{12}}.$$

Here the first term represents twice the expectation value of the one-electron kinetic energy and nuclear attraction (because both electrons occupy ϕ), and the second term is the mean-field electron–electron Coulomb repulsion generated by the charge distribution $|\phi|^2$. The minimization of $E[\phi]$ under the normalization constraint will lead to the Hartree–Fock equation for $\phi(\mathbf{r})$.

(iii) **Variation under normalization constraint using Lagrange multipliers.**

Before proceeding, let us stress that the Lagrange multiplier method is a way to find the minimum of a function of functional subject to a constraint (see Appendix for a little example).

Here the constraint is normalization condition

$$\int d^3r |\phi(\mathbf{r})|^2 = 1.$$

Variation with Lagrange multiplier

We take a Lagrange multiplier ε and vary the functional

$$\mathcal{F}[\phi] = E[\phi] - \varepsilon \left(\int d^3r |\phi(\mathbf{r})|^2 - 1 \right).$$

The idea is to find the function $\phi(\mathbf{r})$ which makes $\mathcal{F}[\phi]$ stationary. In other words, we take a variation of $\mathcal{F}$ with respect to $\phi^*(\mathbf{r})$ and demand that the first-order change vanishes.

Let us define what a functional does. We first consider a single variable $\phi(\mathbf{r})$ assumed real for the moment. Make a small variation of the real function:

$$\phi(\mathbf{r}) \rightarrow \phi(\mathbf{r}) + \delta\phi(\mathbf{r}).$$

The first variation of the functional $F[\phi]$ is defined as

$$\delta F = F[\phi + \delta\phi] - F[\phi].$$

For small variations, the first-order change can always be written in the linear form

$$\delta F = \int d^3r \frac{\delta F}{\delta\phi(\mathbf{r})} \delta\phi(\mathbf{r}).$$

This equation defines the functional derivative $\delta F/\delta\phi(\mathbf{r})$.

Now we go back to our problem, where we have a function and its complex conjugate. In order to find the minimum of the functional, we consider a small change

$$\phi(\mathbf{r}) \rightarrow \phi(\mathbf{r}) + \delta\phi(\mathbf{r}), \quad \phi^*(\mathbf{r}) \rightarrow \phi^*(\mathbf{r}) + \delta\phi^*(\mathbf{r}).$$

We treat ϕ and ϕ^* as independent variables and we want the first-order change $\delta\mathcal{F}$ to vanish for any $\delta\phi^*(\mathbf{r})$ (because of the dependence between the variable and its complex conjugate, minimizing with respect to one suffices). By the definition of functional derivatives (treating ϕ and ϕ^* as independent fields), the first variation of a general functional $F[\phi, \phi^*]$ can be written as

$$\delta F = \int d^3r \left[\frac{\delta F}{\delta\phi(\mathbf{r})} \delta\phi(\mathbf{r}) + \frac{\delta F}{\delta\phi^*(\mathbf{r})} \delta\phi^*(\mathbf{r}) \right].$$

The one-particle energy variation with respect to $\phi^*(\mathbf{r})$ gives We start from the one-electron energy functional

$$E_h[\phi] = 2 \int d^3r \phi^*(\mathbf{r}) \hat{h} \phi(\mathbf{r}),$$

where $\hat{h}$ is a fixed one-particle operator.

The varied energy functional is

$$\begin{aligned} E_h[\phi + \delta\phi] &= 2 \int d^3r (\phi^*(\mathbf{r}) + \delta\phi^*(\mathbf{r})) \hat{h} (\phi(\mathbf{r}) + \delta\phi(\mathbf{r})) \\ &= 2 \int d^3r \left[\phi^*(\mathbf{r}) \hat{h} \phi(\mathbf{r}) + \delta\phi^*(\mathbf{r}) \hat{h} \phi(\mathbf{r}) + \phi^*(\mathbf{r}) \hat{h} \delta\phi(\mathbf{r}) + \delta\phi^*(\mathbf{r}) \hat{h} \delta\phi(\mathbf{r}) \right]. \end{aligned}$$

The first variation is defined as

$$\delta E_h = E_h[\phi + \delta\phi] - E_h[\phi].$$

Substituting the expressions, we obtain

$$\begin{aligned}\delta E_h &= 2 \int d^3r \left[\phi^*(\mathbf{r}) \hat{h} \phi(\mathbf{r}) + \delta\phi^*(\mathbf{r}) \hat{h} \phi(\mathbf{r}) + \phi^*(\mathbf{r}) \hat{h} \delta\phi(\mathbf{r}) + \delta\phi^*(\mathbf{r}) \hat{h} \delta\phi(\mathbf{r}) \right] \\ &\quad - 2 \int d^3r \phi^*(\mathbf{r}) \hat{h} \phi(\mathbf{r}) \\ &= 2 \int d^3r \left[\delta\phi^*(\mathbf{r}) \hat{h} \phi(\mathbf{r}) + \phi^*(\mathbf{r}) \hat{h} \delta\phi(\mathbf{r}) + \delta\phi^*(\mathbf{r}) \hat{h} \delta\phi(\mathbf{r}) \right].\end{aligned}$$

If we keep only the terms linear in the variation (i.e. neglect the quadratic term $\delta\phi^* \hat{h} \delta\phi$), the first-order variation becomes

$$\delta E_h = 2 \int d^3r \left[\delta\phi^*(\mathbf{r}) \hat{h} \phi(\mathbf{r}) + \phi^*(\mathbf{r}) \hat{h} \delta\phi(\mathbf{r}) \right].$$

The derivative is then

$$\frac{\delta E_h}{\delta\phi^*(\mathbf{r})} = 2 \hat{h} \phi(\mathbf{r}).$$

For the electron–electron part we write $|\phi(\mathbf{r}_1)|^2 = \phi^*(\mathbf{r}_1)\phi(\mathbf{r}_1)$ and the same for $\mathbf{r}_2$. The variation with respect to $\phi^*(\mathbf{r})$ has two identical pieces (from the $\mathbf{r}_1$ part and from the $\mathbf{r}_2$ part). Collecting these two contributions we obtain

$$\frac{\delta E_{12}}{\delta\phi^*(\mathbf{r})} = 2 \phi(\mathbf{r}) \int d^3r' \frac{e^2}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|} |\phi(\mathbf{r}')|^2.$$

It is convenient to define the Hartree potential

$$V_H(\mathbf{r}) = \int d^3r' \frac{e^2}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|} |\phi(\mathbf{r}')|^2,$$

so that the Coulomb contribution to the Euler–Lagrange equation reads

$$\frac{\delta E_{12}}{\delta\phi^*(\mathbf{r})} = 2 V_H(\mathbf{r}) \phi(\mathbf{r}).$$

We consider the constraint term and first notice that the constant term $+\varepsilon$ has zero variation. The integrand's variation is

$$\begin{aligned}\delta(\phi^* \phi) &= (\phi^* + \delta\phi^*)(\phi + \delta\phi) - \phi^* \phi \\ &= \phi^* \phi + \phi^* \delta\phi + \delta\phi^* \phi + \delta\phi^* \delta\phi - \phi^* \phi \\ &\simeq \phi^*(\mathbf{r}) \delta\phi(\mathbf{r}) + \delta\phi^*(\mathbf{r}) \phi(\mathbf{r}),\end{aligned}$$

where in the last step we have dropped the second–order term $\delta\phi^* \delta\phi$.

Therefore the variation of $\mathcal{F}'$ is

$$-\varepsilon \int d^3r \delta[\phi^*(\mathbf{r})\phi(\mathbf{r})] = -\varepsilon \int d^3r \left[\phi^*(\mathbf{r}) \delta\phi(\mathbf{r}) + \delta\phi^*(\mathbf{r}) \phi(\mathbf{r}) \right].$$

Comparing this with the explicit expression

$$-\varepsilon \int d^3r \phi^*(\mathbf{r}) \delta\phi(\mathbf{r}) - \varepsilon \int d^3r \delta\phi^*(\mathbf{r}) \phi(\mathbf{r}),$$

we can read off the functional derivatives as the coefficients of the independent variations $\delta\phi$ and $\delta\phi^*$:

$$\begin{aligned}\frac{\delta\mathcal{F}'}{\delta\phi(\mathbf{r})} &= -\varepsilon\phi^*(\mathbf{r}), \\ \frac{\delta\mathcal{F}'}{\delta\phi^*(\mathbf{r})} &= -\varepsilon\phi(\mathbf{r}).\end{aligned}$$

Restoring the constant term (which does not affect the derivatives), we obtain for the original normalization functional

$$\mathcal{F}_{\text{norm}}[\phi, \phi^*] = -\varepsilon \left(\int d^3r |\phi(\mathbf{r})|^2 - 1 \right),$$

the result

$$\frac{\delta}{\delta\phi^*(\mathbf{r})} \left[-\varepsilon \left(\int d^3r |\phi(\mathbf{r})|^2 - 1 \right) \right] = -\varepsilon\phi(\mathbf{r}).$$

Collecting all contributions we have

$$\left[2\hat{h} + 2V_{\text{H}}(\mathbf{r}) - \varepsilon \right] \phi(\mathbf{r}) = 0.$$

Dividing by 2 and defining $\varepsilon' = \varepsilon/2$ gives the Hartree–Fock (here Hartree) equation

$$\left[\hat{h} + V_{\text{H}}(\mathbf{r}) \right] \phi(\mathbf{r}) = \varepsilon' \phi(\mathbf{r}).$$

We have therefore converted the original two–electron Schrödinger equation into a self–consistent one–electron equation where the potential includes the average Coulomb field produced by the second electron.

(iv) **Self-consistency loop.**

In practice one solves this equation self-consistently: start from a trial orbital $\phi^{(0)}(\mathbf{r})$, compute the Coulomb (mean-field) potential from $|\phi^{(0)}|^2$, solve the resulting single-particle equation for an improved orbital $\phi^{(1)}$, and iterate until the orbital and energy no longer change within the desired accuracy.

Example of computing the helium ground state

While the general application of this theory is fairly involved, we now circumvent the full Hartree Fock procedure by using a simplified orbital with only a free parameter Z :

$$\phi_Z(\mathbf{r}) = \frac{Z^{3/2}}{\sqrt{\pi a_0^3}} e^{-Zr/a_0}.$$

This parameter Z will be determined variationally from minimizing the energy but without making use of the Lagrange multiplier as the normalization condition is fulfilled automatically at any point. This shortcut is allowed as our energy functional turns into a very simple function of Z .

By definition,

$$\left\langle \frac{1}{r} \right\rangle = \int d^3r |\phi_Z(\mathbf{r})|^2 \frac{1}{r}.$$

Using spherical coordinates $d^3r = r^2 \sin\theta dr d\theta d\varphi$, and noting that $|\phi_Z|$ depends only on r , we obtain

$$\left\langle \frac{1}{r} \right\rangle = 4\pi \int_0^\infty dr r |\phi_Z(r)|^2 = 4\pi \left(\frac{Z^3}{\pi a_0^3} \right) \int_0^\infty dr r e^{-2Zr/a_0} = \frac{Z}{a_0}.$$

For the expectation value of the kinetic energy, we want

$$\left\langle -\frac{\hbar^2}{2m_e} \nabla^2 \right\rangle = -\frac{\hbar^2}{2m_e} \int d^3r \phi_{1s}^*(\mathbf{r}) \nabla^2 \phi_Z(\mathbf{r}) = \frac{\hbar^2 Z^2}{2m_e a_0^2}.$$

Therefore, the total one-electron contribution is

$$E_h(Z) = 2 \langle \hat{h}_{\text{He}} \rangle = \frac{\hbar^2 Z^2}{m_e a_0^2} - \frac{2e^2}{2\pi\epsilon_0} \frac{Z}{a_0}.$$

The electron-electron energy in Hartree theory is

$$E_{12}(Z) = \iint d^3r_1 d^3r_2 \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|} |\phi_Z(\mathbf{r}_1)|^2 |\phi_Z(\mathbf{r}_2)|^2.$$

Using the explicit form $|\phi_Z(\mathbf{r})|^2 = \frac{Z^3}{\pi a_0^3} e^{-2Zr/a_0}$, this integral can be carried out analytically and gives the well-known Coulomb integral for two identical $1s$ orbitals:

$$E_{12}(Z) = \frac{5}{8} \frac{e^2}{4\pi\epsilon_0} \frac{Z}{a_0}.$$

Adding the two contributions gives

$$E(Z) = \frac{\hbar^2 Z^2}{m_e a_0^2} - \frac{2e^2}{2\pi\epsilon_0} \frac{Z}{a_0} + \frac{5}{8} \frac{e^2}{4\pi\epsilon_0} \frac{Z}{a_0} = \frac{\hbar^2 Z^2}{m_e a_0^2} - \frac{27}{8} \frac{e^2}{4\pi\epsilon_0} \frac{Z}{a_0}.$$

This expression represents the Hartree approximation to the helium ground-state energy for a trial orbital labeled by Z . To find the minimum, we differentiate with respect to Z :

$$\frac{dE}{dZ} = \frac{2\hbar^2 Z}{m_e a_0^2} - \frac{27}{8} \frac{e^2}{4\pi\epsilon_0 a_0}.$$

Using the definition of the Bohr radius,

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} \Rightarrow \frac{e^2}{4\pi\epsilon_0 a_0} = \frac{\hbar^2}{m_e a_0^2},$$

we can rewrite

$$\frac{dE}{dZ} = \frac{\hbar^2}{m_e a_0^2} \left(2Z - \frac{27}{8} \right).$$

The minimum satisfies $\frac{dE}{dZ} = 0$, hence

$$2Z - \frac{27}{8} = 0 \Rightarrow Z_{\min} = \frac{27}{16}.$$

At this value,

$$E_{\min} = E\left(\frac{27}{16}\right) = \frac{e^2}{4\pi\epsilon_0 a_0} \left[\left(\frac{27}{16}\right)^2 - \frac{27}{8} \left(\frac{27}{16}\right) \right] = -\frac{729}{256} \frac{e^2}{4\pi\epsilon_0 a_0}.$$

Substituting Z_{opt} back into $E(Z)$ gives

$$E_{\min} = E(Z_{\text{opt}}) \approx -2.85 \text{ Ha},$$

which is already quite close to the exact nonrelativistic helium ground-state energy

$$E_{\text{exact}} \approx -2.903 \text{ Ha}.$$

Although we used only a single variational parameter, the Hartree procedure already captures most of the binding energy and produces a physically reasonable screening of the nuclear charge. In a full self-consistent Hartree-Fock calculation the orbital shape itself would also be updated iteratively, but the present simple ansatz already illustrates the mechanism: the orbital generates a mean Coulomb potential, and the optimal orbital is the one that is consistent with this potential.

Appendix: Lagrange multiplier method

Lagrange multipliers (simple example)

To minimize a function $f(x, y)$ with the constraint $g(x, y) = 1$, we introduce an additional function

$$F(x, y) = f(x, y) - \lambda(g(x, y) - 1),$$

which has the constraint built in. The minimum is found by equating and solve

$$\frac{\partial F}{\partial x} = 0, \quad \frac{\partial F}{\partial y} = 0,$$

and enforcing that , $g(x, y) = 1$. The multiplier λ enforces the constraint.

Example: Minimize $f(x, y) = x^2 + y^2$ subject to $x + y = 1$.

$$F(x, y) = x^2 + y^2 - \lambda(x + y - 1).$$

Stationarity:

$$\frac{\partial F}{\partial x} = 2x - \lambda = 0, \quad \frac{\partial F}{\partial y} = 2y - \lambda = 0 \Rightarrow 2x = \lambda, \quad 2y = \lambda \Rightarrow x = y.$$

Constraint $x + y = 1$ gives $x = y = \frac{1}{2}$, so the minimum occurs at $(x, y) = (\frac{1}{2}, \frac{1}{2})$.

Appendix: Functional calculus

Definition of the variation δF

For any functional $F[\phi]$, the *variation* is defined as

$$\delta F = F[\phi + \delta\phi] - F[\phi].$$

Expanding to first order in $\delta\phi$ gives the linear response of the functional under a small change in the field. For example, for

$$F[\phi] = \int d^3r \frac{1}{r^2} \phi(\mathbf{r}),$$

we compute:

$$F[\phi + \delta\phi] = \int d^3r \frac{1}{r^2} (\phi(\mathbf{r}) + \delta\phi(\mathbf{r})) = F[\phi] + \int d^3r \frac{1}{r^2} \delta\phi(\mathbf{r}).$$

Therefore,

$$\delta F = F[\phi + \delta\phi] - F[\phi] = \int d^3r \frac{1}{r^2} \delta\phi(\mathbf{r}).$$

By the definition of the functional derivative,

$$\delta F = \int d^3r \frac{\delta F}{\delta\phi(\mathbf{r})} \delta\phi(\mathbf{r}),$$

so we identify

$$\frac{\delta F}{\delta\phi(\mathbf{r})} = \frac{1}{r^2}.$$

Appendix: Computing the energy with the variable Z orbital

Using spherical coordinates $d^3r = r^2 \sin \theta dr d\theta d\varphi$, and noting that $|\phi_Z|$ depends only on r , we obtain

$$\left\langle \frac{1}{r} \right\rangle = 4\pi \int_0^\infty dr r |\phi_Z(r)|^2 = 4\pi \left(\frac{Z^3}{\pi a_0^3} \right) \int_0^\infty dr r e^{-2Zr/a_0}.$$

Set

$$\alpha = \frac{2Z}{a_0}, \quad \int_0^\infty r e^{-\alpha r} dr = \frac{1}{\alpha^2},$$

so that

$$\int_0^\infty r e^{-2Zr/a_0} dr = \frac{a_0^2}{4Z^2}.$$

Therefore

$$\left\langle \frac{1}{r} \right\rangle = 4\pi \left(\frac{Z^3}{\pi a_0^3} \right) \frac{a_0^2}{4Z^2} = \frac{Z}{a_0}.$$

For the expectation value of the kinetic energy, we want

$$\left\langle -\frac{\hbar^2}{2m_e} \nabla^2 \right\rangle = -\frac{\hbar^2}{2m_e} \int d^3r \phi_{1s}^*(\mathbf{r}) \nabla^2 \phi_Z(\mathbf{r}).$$

Write

$$\phi_Z(r) = A e^{-\lambda r}, \quad A = \left(\frac{Z^3}{\pi a_0^3} \right)^{1/2}, \quad \lambda = \frac{Z}{a_0}.$$

As no angular dependence is present,

$$\nabla^2 \phi(r) = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right).$$

Compute derivatives:

$$\frac{d\phi}{dr} = -\lambda A e^{-\lambda r}, \quad r^2 \frac{d\phi}{dr} = -\lambda A r^2 e^{-\lambda r}, \quad \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = -\lambda A e^{-\lambda r} (2r - \lambda r^2).$$

Thus

$$\nabla^2 \phi(r) = \frac{-\lambda A e^{-\lambda r} (2r - \lambda r^2)}{r^2} = -\lambda A e^{-\lambda r} \left(\frac{2}{r} - \lambda \right).$$

Since $\phi(r) = A e^{-\lambda r}$, this becomes

$$\nabla^2 \phi_Z(r) = \left(\lambda^2 - \frac{2\lambda}{r} \right) \phi_Z(r).$$

Hence

$$\langle \nabla^2 \rangle = \int d^3r |\phi_Z|^2 \left(\lambda^2 - \frac{2\lambda}{r} \right).$$

Split the terms:

$$\int d^3r |\phi_Z|^2 = 1, \quad \int d^3r |\phi_Z|^2 \frac{1}{r} = \frac{Z}{a_0} = \lambda.$$

Therefore

$$\langle \nabla^2 \rangle = \lambda^2 - 2\lambda(\lambda) = -\lambda^2.$$

Thus the kinetic expectation value is

$$\left\langle -\frac{\hbar^2}{2m_e} \nabla^2 \right\rangle = -\frac{\hbar^2}{2m_e} (-\lambda^2) = \frac{\hbar^2 \lambda^2}{2m_e} = \frac{\hbar^2 Z^2}{2m_e a_0^2}.$$

Therefore, the total one-electron contribution is

$$E_h(Z) = 2 \langle \hat{h}_{\text{He}} \rangle = \frac{\hbar^2 Z^2}{m_e a_0^2} - \frac{2e^2}{2\pi\epsilon_0} \frac{Z}{a_0}.$$

The electron-electron energy in Hartree theory is

$$E_{12}(Z) = \iint d^3r_1 d^3r_2 \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|} |\phi_Z(\mathbf{r}_1)|^2 |\phi_Z(\mathbf{r}_2)|^2.$$

Using the explicit form $|\phi_Z(\mathbf{r})|^2 = \frac{Z^3}{\pi a_0^3} e^{-2Zr/a_0}$, this integral can be carried out analytically and gives the well-known Coulomb integral for two identical $1s$ orbitals:

$$E_{12}(Z) = \frac{5}{8} \frac{e^2}{4\pi\epsilon_0} \frac{Z}{a_0}.$$